

LIDAR-based windthrow risk mapping in a German deciduous forest

1 Introduction

For predicting the windthrow risk of trees and forests, a broad variety of different empirical and mechanical models can be utilized (Gardiner, et al., 2008). While many of these algorithms are taking local wind climate and airflow into account, others are based specifically on observable and measurable tree properties to model the stress imposed on the vegetation caused by wind loading. For assessing windthrow risk predicted by mechanical models, usually a two-stages workflow is initialized, consisting of calculating the above-canopy critical windspeed (CWS) as first step and taking the local wind climatology into account as second step (Gardiner, et al., 2008).

In this study, only the first part of the modelling is applied, to focus mainly on tree attributes which can be estimated by analysing a LIDAR point cloud generated over a forest. As a result, the critical windspeed for breaking and overturning, respectively, will be approximated for each segmented tree.

2 Study Site and Data

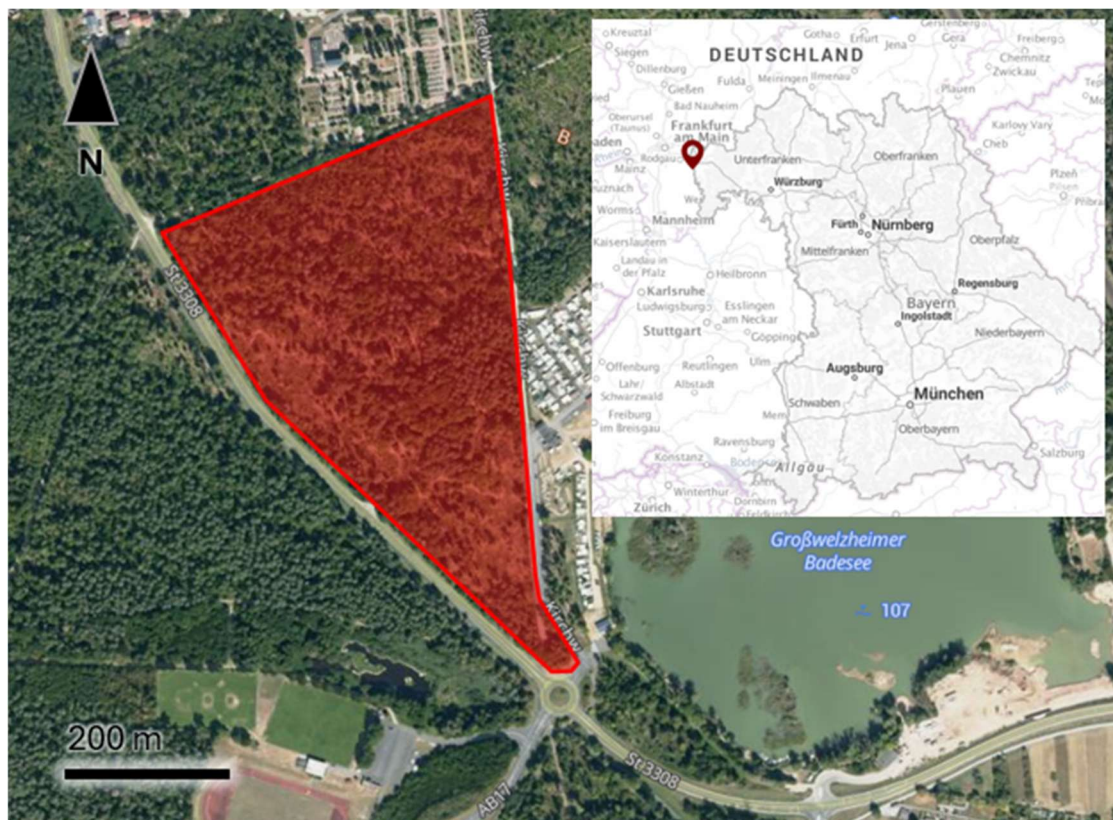


Figure 1: Study site in northwestern Bavaria (LDBV Bayern, 2026)

The study site is located in the very northwest of Bavaria, Germany at approx. 50.058 °N and 9.012 °E. Being close to the Spessart mountain range as one of the largest connected deciduous forests in Germany, the area is characterized by similar vegetation, consisting mainly of oak and beech trees.

A LIDAR-based point cloud covering the area was downloaded from the geodata service of the Bavaria state surveillance program, where LIDAR datasets as 1 sqkm-tiles are freely provided. Additionally, a Digital Elevation Model (DEM) for the area of interest (AOI) was obtained from the same site as well (LDBV Bayern, 2026).

3 Methods

There are two mathematical indicators referring to the CWS; the first one being the CWS at canopy top for stem breakage (CWS_break) and the latter one describing the CWS at canopy top for overturning (CWS_overturn) (Gardiner, et al., 2008).

$$CWS_{break} = \frac{1}{kD} \left[\frac{\pi MOR \times d.b.h.^3}{32 \rho G (d-1.3)} \right]^{\frac{1}{2}} \left[\frac{f_{knot}}{f_{edge} f_{CW}} \right]^{\frac{1}{2}} \ln \left(\frac{h-d}{z_0} \right), \quad (1)$$

$$CWS_{over} = \frac{1}{kD} \left[\frac{C_{reg} SW}{\rho G d} \right]^{\frac{1}{2}} \left[\frac{1}{f_{edge} f_{CW}} \right]^{\frac{1}{2}} \ln \left(\frac{h-d}{z_0} \right), \quad (2)$$

The following table lists the necessary parameters along with their sources:

Parameter	Meaning	Value retrieved by
k	Karman constant	literature
D	average tree spacing [m]	LIDAR point cloud
MOR	modulus of rupture [Pa]	literature
d.b.h.	diameter at breast height [m]	LIDAR point cloud
p	air density [kg/m-3]	literature
G	empirical gust factor	literature
d	zero-plane displacement [m]	LIDAR point cloud
f_knot	wood strength reduction factor	literature
f_edge	wood strength reduction factor	literature
f_CW	wood strength reduction factor	literature
h	tree height [m]	LIDAR point cloud
z_0	aerodynamic roughness [m]	LIDAR point cloud
C_reg	regression constant	literature
SW	stem weight [kg]	LIDAR point cloud

The workflow to retrieve the tree-dependent CWS was separated into four stages:

1 Preliminary Tasks

First, both the point cloud and DEM were clipped to the extent of the AOI using QGIS. Next, the point cloud was imported in R and classified in ground- and non-ground points before additionally, the external DEM was subtracted from the classified points. The resulting normalized point cloud was then filtered to only keep points above 0.4 in height, due to low-growing canopy, which the succeeding formulas are not suited to be applied to. A canopy height model was created, using the *rasterize_canopy()* algorithm from the lidR package, which was then smoothed using a focal lens with a kernel size of [1,5,5] (Roussel, Goodbody, & Tompalski, 2026).

To be able to concentrate the analysis on the respective trees, the tree tops were located by the *locate_trees()* function and later used as an argument within the *segment_trees()* function, where the Dalponte2016 algorithm was applied (Roussel, Goodbody, & Tompalski, 2026). The result is a segmented point cloud, with each tree clearly distinguishable.

2 Parameter Retrieval

A) Average Spacing D between Trees

To get a single spacing value for each tree instead of an universal value dedicated to the whole AOI, the parameter was interpreted as the mean distance between each reference tree and its respective neighbours. By using the *st_is_within_distance()* function (Pebesma, 2026) and the extraction of each neighbour's coordinates, connecting lines were constructed to measure the distances. The resulting values were added to the treetops-data frame.

B) Aerodynamic Profile and Tree Height

To approximate the aerodynamic profile of each tree, the spatial extent of every segment in X-, Y- and Z-direction as well as its number of points inside was calculated to account for the local point density.

$$\text{Leaf Area Index, LAI} = \frac{n_p d(x_i + y_i)}{x_i y_i} \left(\frac{1}{z_i} \right) \quad (3)$$

Next, the Leaf Area Index (LAI) was estimated by relying on a point spatial density algorithm proposed by Indirabai et al., 2020, which needs the number of points n_p , the dimensions of each tree's bounding box and the point spacing d of the cloud, whereat the latter can be derived directly from the point cloud statistics.

The aerodynamic roughness for each tree can be predicted by the Raupach model (Raupach, 1992), which is described in Floors et al., 2021.

$$\lambda = 0.5 \text{LAI}. \quad (4)$$

$$a = \sqrt{2 \cdot 7.5 \cdot \lambda} \quad (5)$$

$$\frac{d}{h} = 1 - \frac{1 - \exp(-a)}{a} \quad (6)$$

$$\frac{z_0}{h} = b \exp \left(\frac{-\kappa}{\min(\sqrt{C_S + C_R \lambda}, c_{\max})} - \Psi_h \right) \quad (7)$$

Aside from the Karman constant ($\kappa = 0.4$), the following parameters were used with empirical values suggested by Raupach, 1994: $C_S = 0.003$, $C_R = 0.3$, $c_{\max} = 0.3$, and $\Psi_h = 0.193$.

C) Diameter at Breast Height (DBH)

Due to the airborne acquisition of the LIDAR point cloud and the rather low point density, when compared to data obtained from the ground, the position size of each tree stem had to be approximated first.

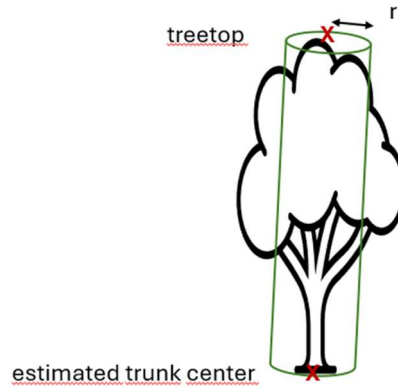


Figure 2: Constructing a cylindric cone over each tree

Therefore, the segmented point cloud was filtered, keeping only the points above 0.4 m and below the breast height of 1.3 m above ground. The resulting points of each segment were assumed as tree trunk; and center point was calculated depicting the average XY-coordinates of the inlying points. A line was then constructed, spanning from the trunk center to the tree top to place a projected cylindric cone over the tree. For the starting radius, a value of 0.5 was used, to cut off most of the tree crown.

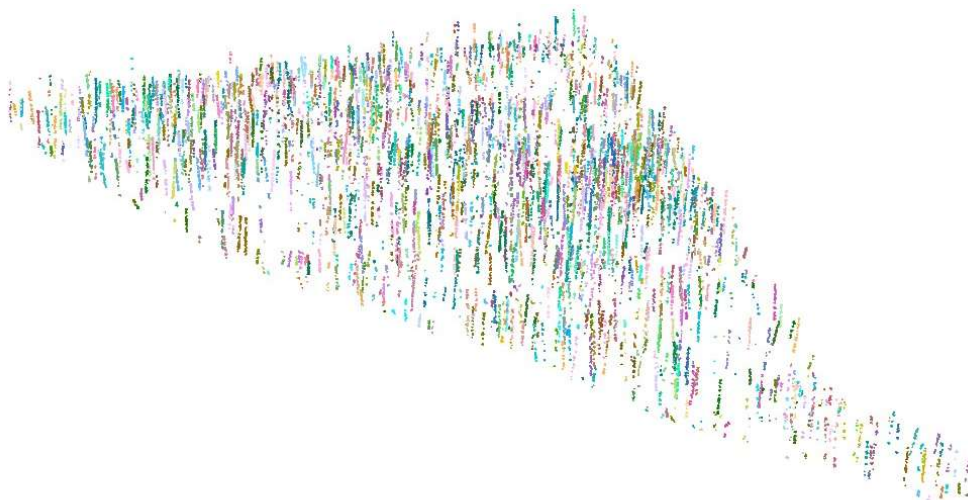


Figure 3: Remaining stem segments after the first filtering with $r = 0.5$

The resulting stem segments were then filtered into four sections along the vertical axis, before in a next step, a 2D horizontal circle was fitted onto the points of each segment, using the *lidR::fit_circle()* function (Roussel, Goodbody, & Tompalski, 2026). For each tree segment, the resulting circle radii of the respective sections were then averaged to receive a new, tree-dependent starting radius for a second iteration of the cylindric cone fitting.

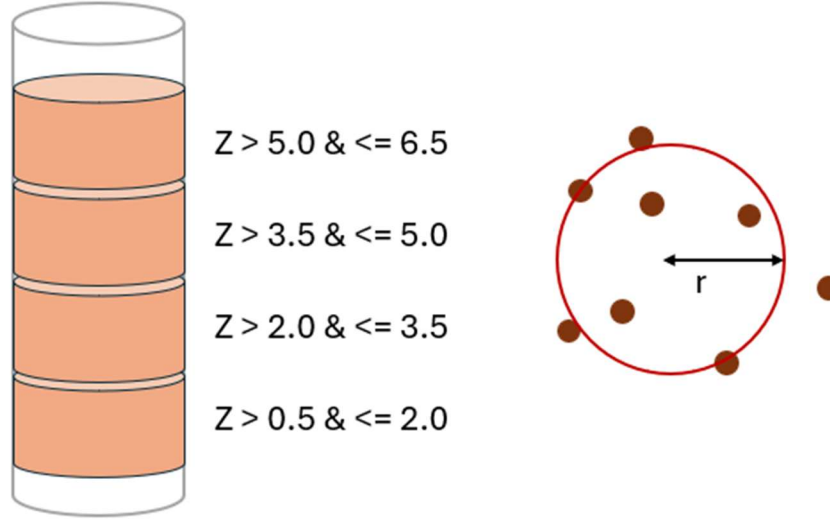


Figure 4: Fitting a horizontal circle on each stem section

After the second estimation of the tree stems, each stem segment was filtered again to keep a vertical section ranging from 1.3-0.5 to 1.3+0.5 at approximate breast height. Here, the circle fitting was applied again to receive the radius, and in the following, the diameter at breast height.

Since the explained method for DBH retrieval was not successful for all the trees, an additional method was applied to the remaining segments, following an approach from Mhlaba Bhebhe, Liu, & Paudyal, 2025, which uses the tree height [m] to fit a curve.

$$DBH = a_1 \times (H - 1.3)a_2 + \varepsilon_i \quad (8)$$

D) Stem Weight estimation

As described in the previous sections, the tree stems are hardly delineable; hence an approach for estimating the biomass of standing trees was used, described in Mhlaba Bhebhe, Liu, & Paudyal, 2025. The algorithm needs DBH [m], tree height H [m] and a specific wood density value [g/cm³] as inputs.

$$(AGB)_{est} = \exp(-2.187 + 0.916 \times \ln(\rho D^2 H)) \quad (9)$$

Since the trees within the AOI are mainly a composite of oak and beech trees, the respective density values (0.6 and 0.9) were taken from Engineeringtoolbox, 2026, averaged and used for p.

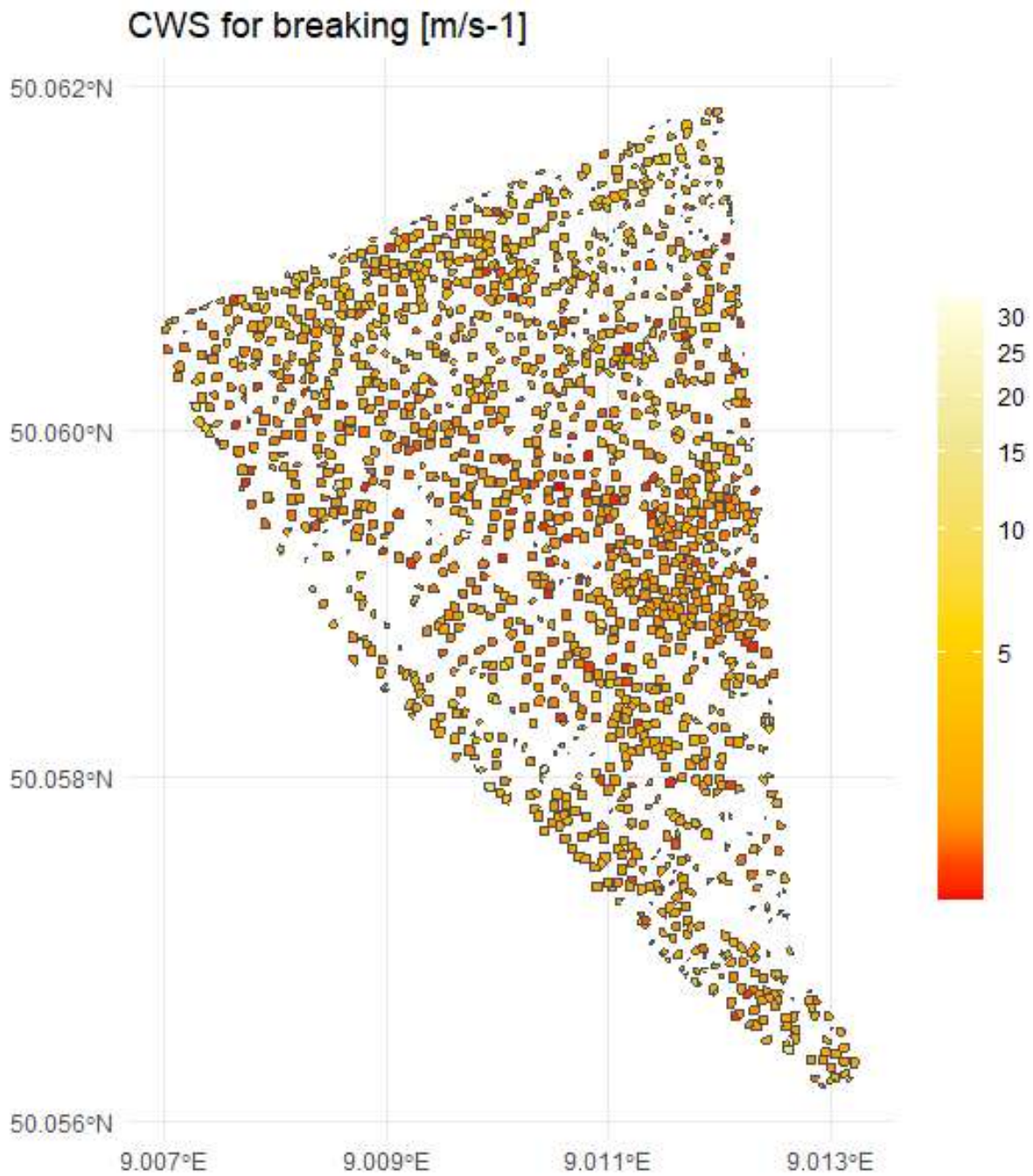
3 Final CWS estimation

With all LIDAR-derived parameters added to the dataframe, the final calculation for CWS_break and CWS_overturn was made, using formulas (1) and (2).

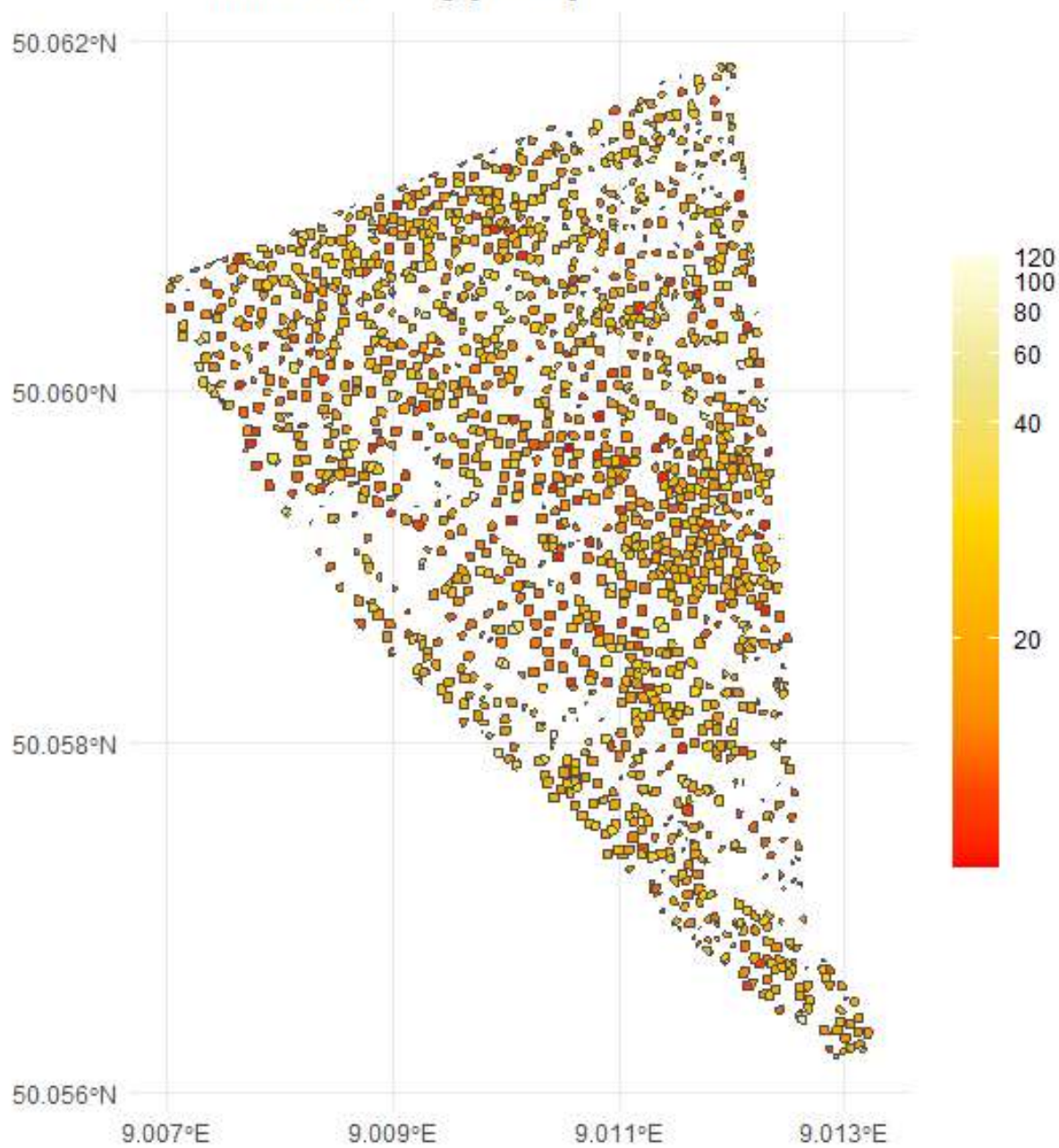
In addition, the remaining parameters needed for the two calculations, were retrieved from literature. The following table lists their respective values and sources:

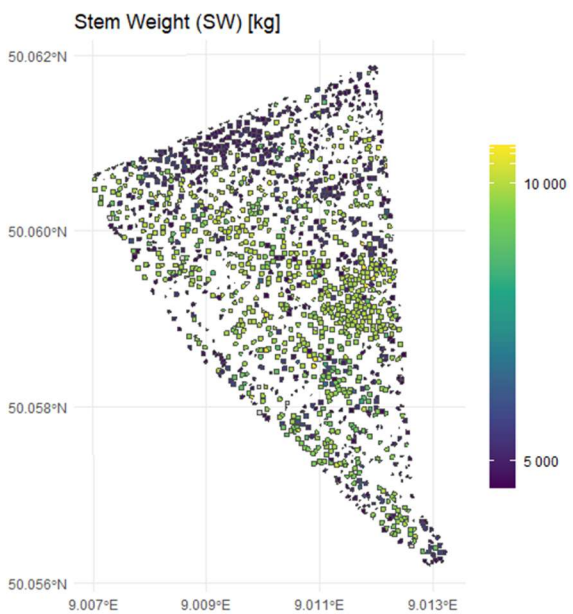
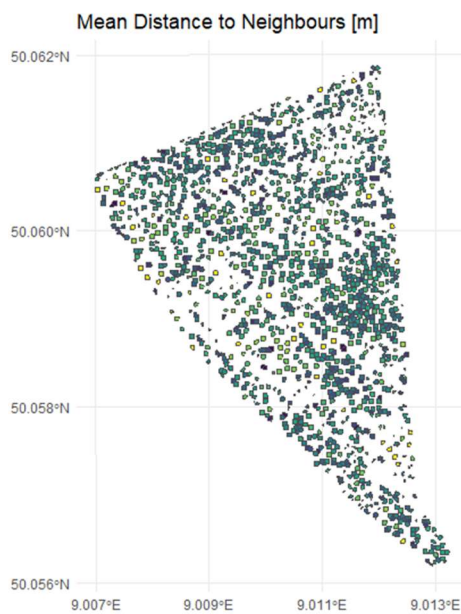
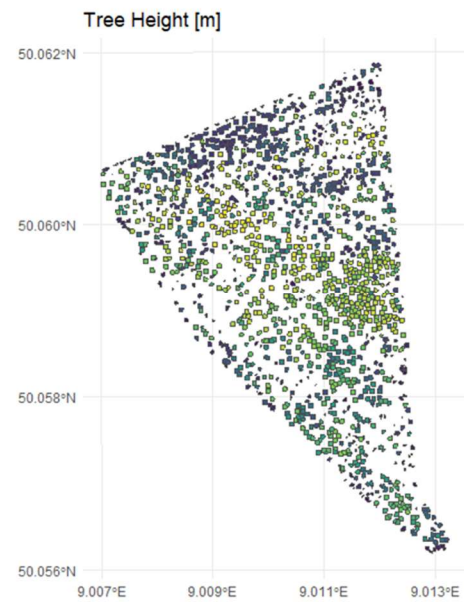
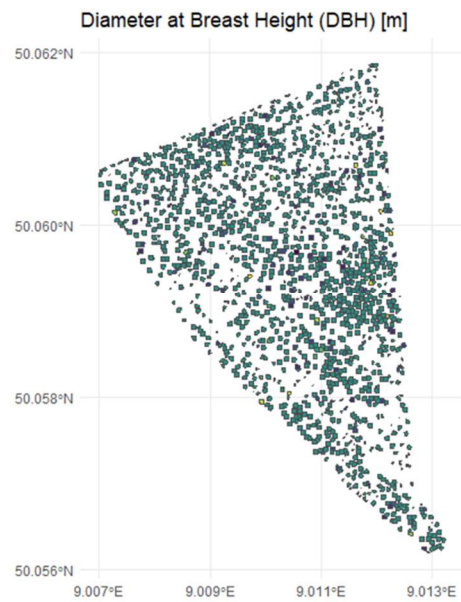
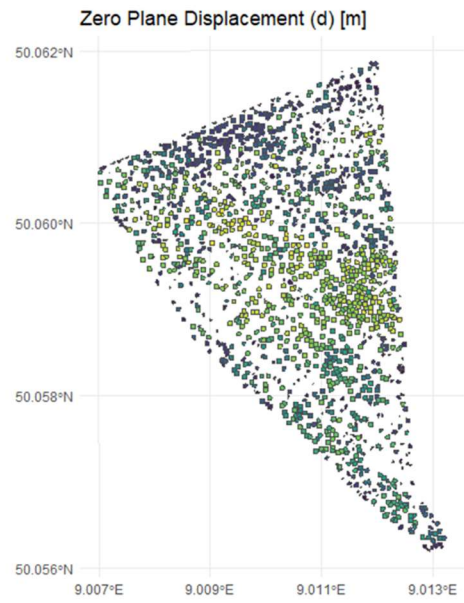
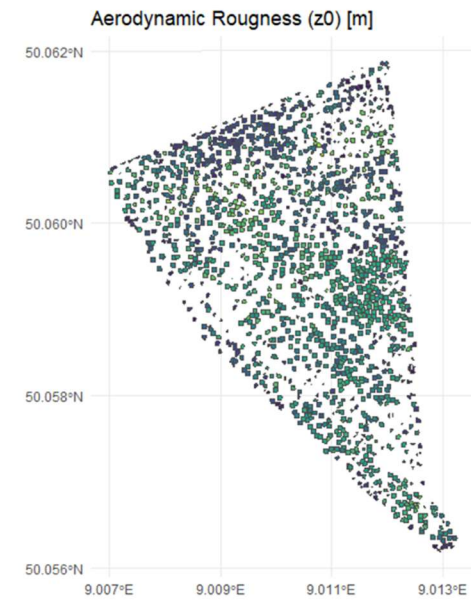
Parameter	Value used	Reference
k	0.4	(Gardiner, et al., 2008)
MOR	105000	(Kumar, 2026)
G	2.0	(Gross, 2018)
p	1.2 kg / m ³	(Density Calculator, 2026)
C_{reg}	190	(Nicoll, Gardiner, Rayner,, & Peace, 2006)
f_{knot}	<i>assumed to be 1</i>	(Gardiner, et al., 2008)
f_{edge}	<i>assumed to be 1</i>	(Gardiner, et al., 2008)
f_{cw}	<i>assumed to be 1</i>	(Gardiner, et al., 2008)

4 Results



CWS for overturning [m/s-1]





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